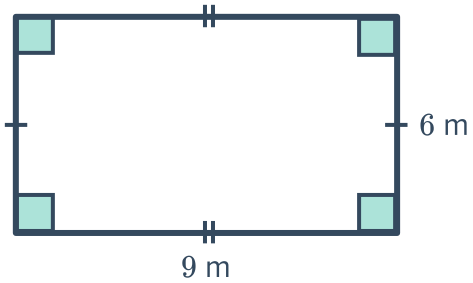
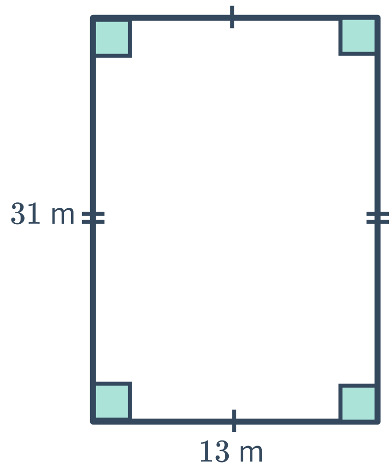
Areas of rectangles, squares and tri[redacted]es

1 Find the area of the following rectangles:

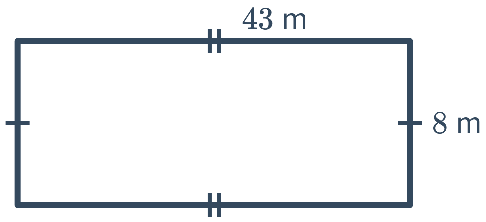
a



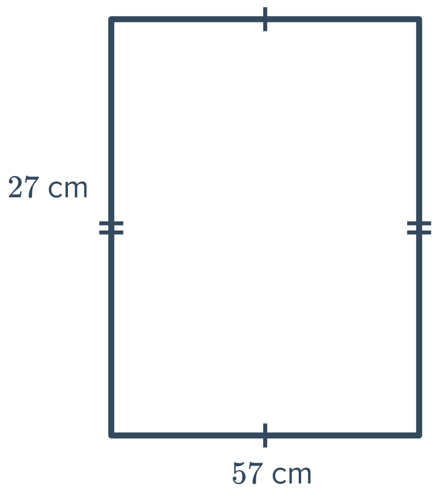
b



c



d



2 Find the area of a rectangle whose length is 12 cm and width is 5 cm.

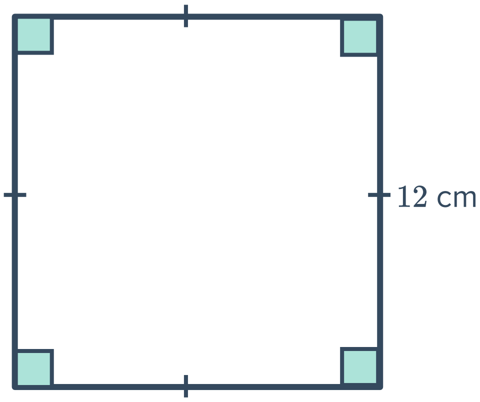
3 Find the area of a rectangle whose length is 11 m and width is 8 m.

4 If the area of a rectangle is 99 cm^2 and its width is 9 cm, find its length.

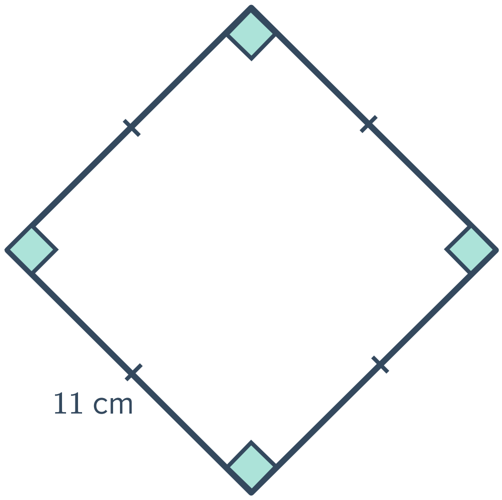
5 If the area of a rectangle is 160 cm^2 and its length is 20 cm, find its width.

6 Find the area of the following squares:

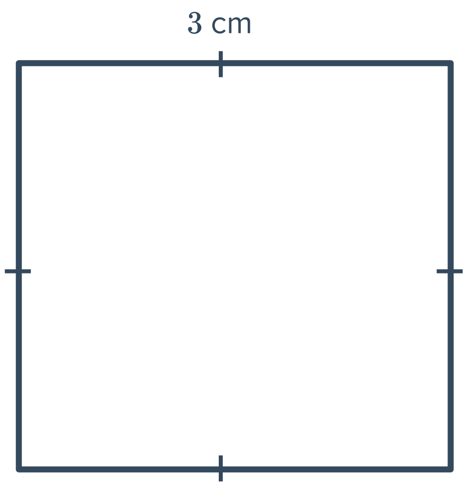
a



b



c



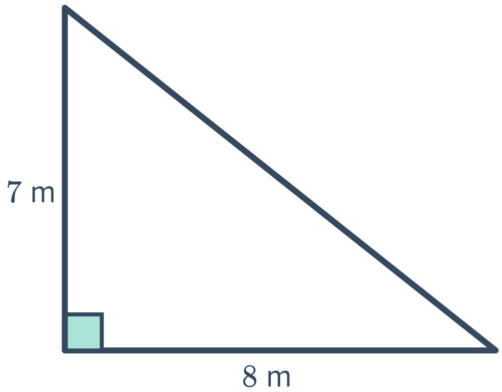
7 Find the length of each side of a square whose area is:

a 64 cm^2

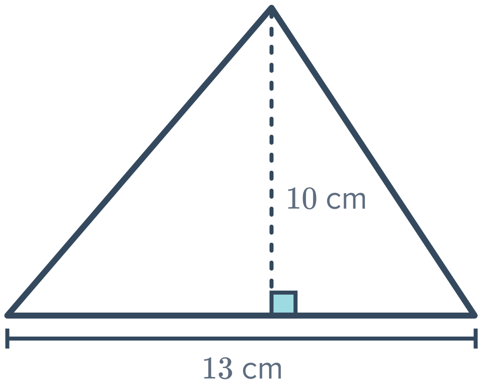
b 36 m^2

8 Find the area of the following triangles:

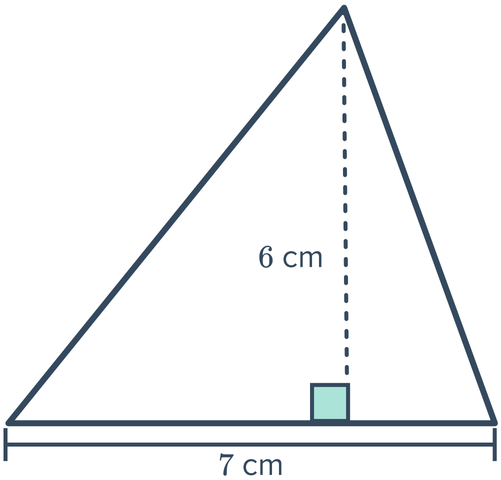
a



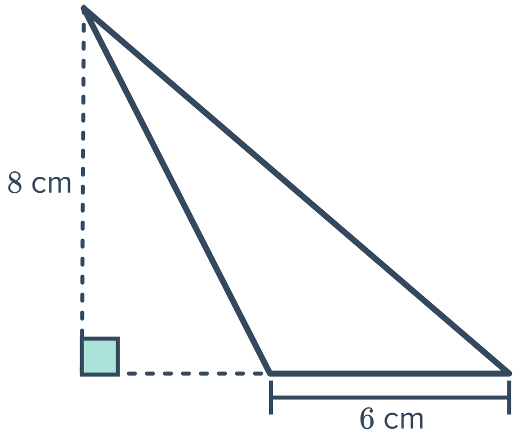
b



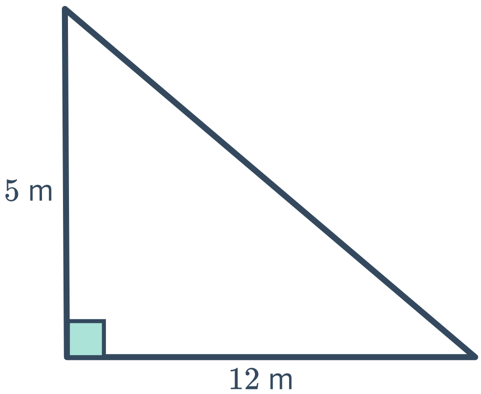
c



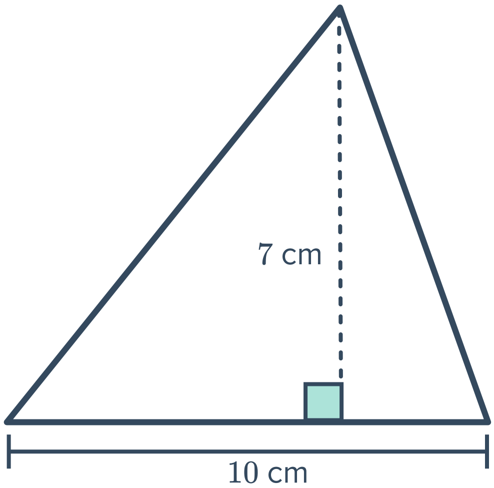
d



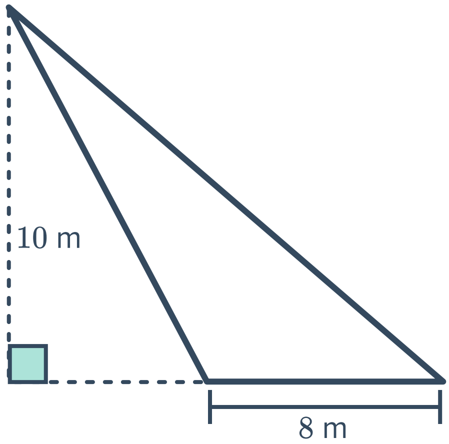
e



f

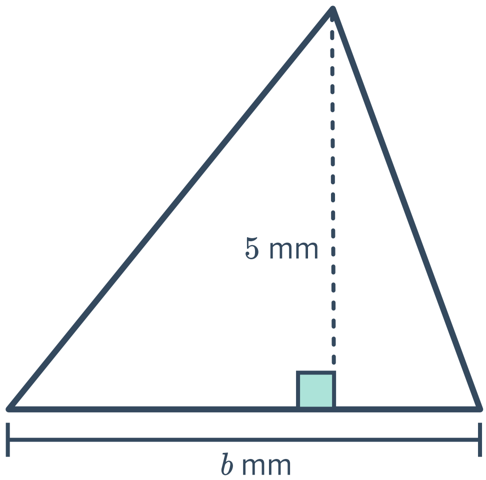


g

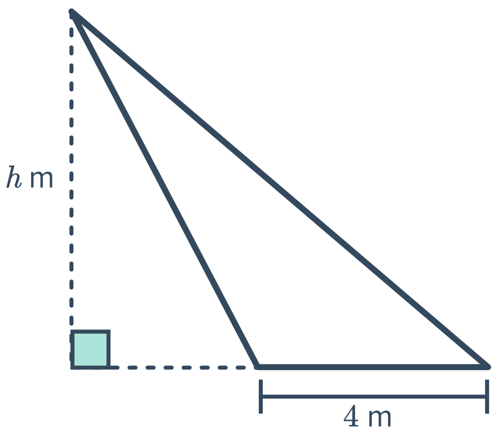


9 Find the value of the pronumeral for each of the following triangles, given the area:

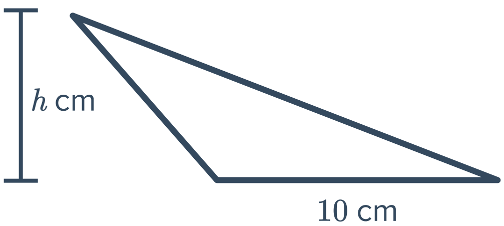
a The area of this triangle is $20\ \mathrm{mm}^2$



b The area of this triangle is $48\ \mathrm{m}^2$.



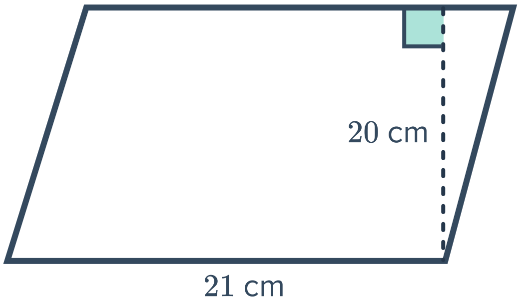
c The area of this triangle is $130\ \mathrm{cm}^2$.



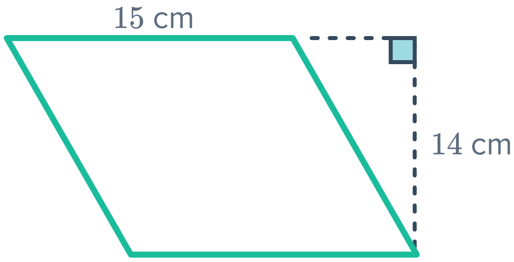
Areas of parallelograms

11 Find the area of the following parallelograms:

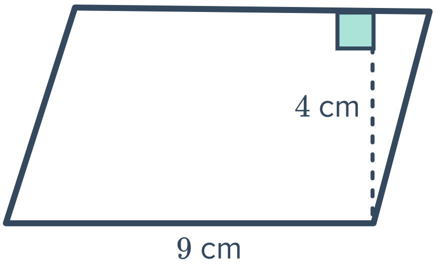
a



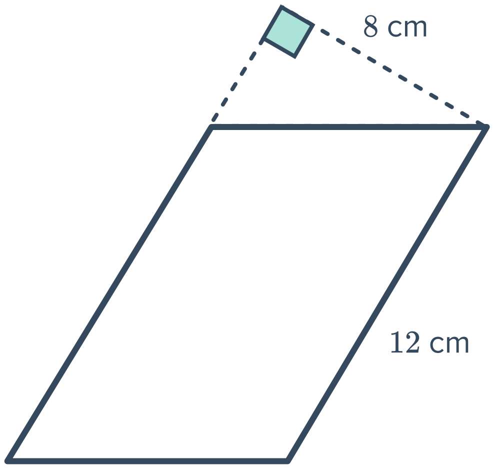
b



c



d



12 Find the area of a parallelogram with:

a A base of 5 mm and perpendicular height of 2 mm.

b A base of 15 cm and a perpendicular height of 7 cm.

c A base of 10 cm and a perpendicular height 5 cm.

13 Find the base length of a parallelogram with area of 96 cm^2 and perpendicular height of 8 cm.

14 Find the height of a parallelogram with area of 84 cm^2 and perpendicular base length of 6 cm

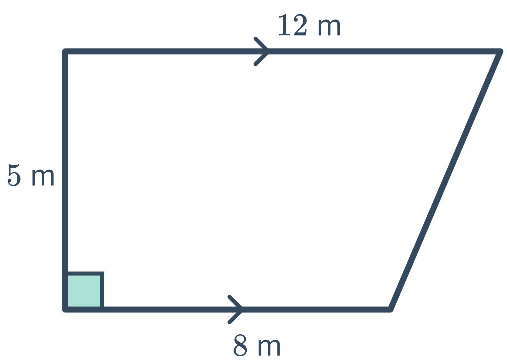
15 Find the perpendicular height, h, of a parallelogram that has an area of $45\,\text{cm}^2$ and a base length of $5\,\text{cm}$.

16 Find the base length, b, of a parallelogram that has an area of $216\,\text{mm}^2$ and a perpendicular height of $12\,\text{mm}$.

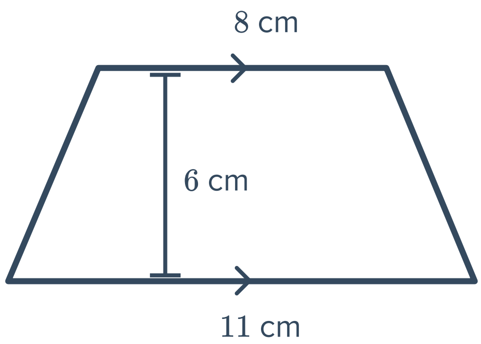
Areas of trapeziums

17 Find the area of the following trapeziums:

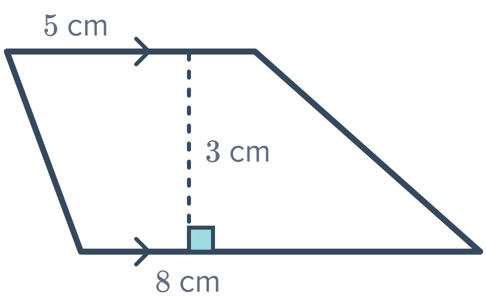
a



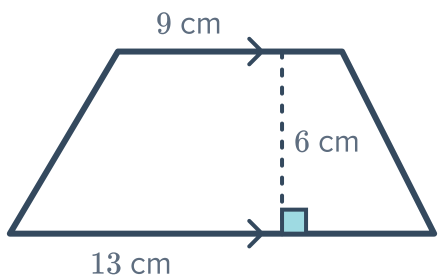
b



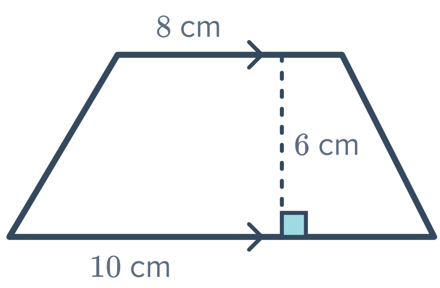
c



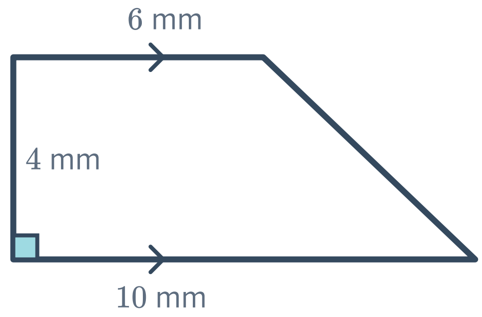
d



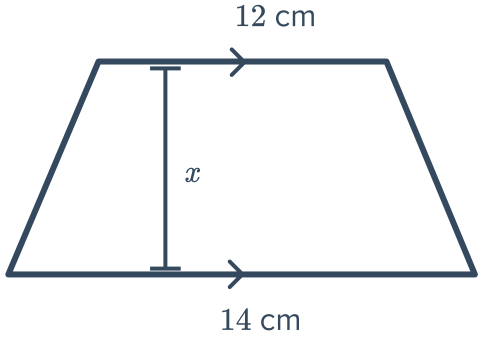
e



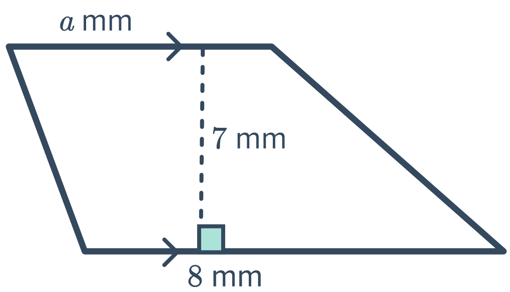
f



18 Find the value of x if the area of the trapezium shown is $65\ \mathrm{cm}^2$.



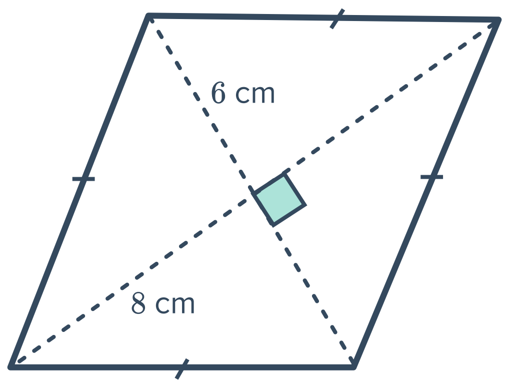
19 Find the value of a if the area of the trapezium is $42\ \mathrm{mm}^2$.



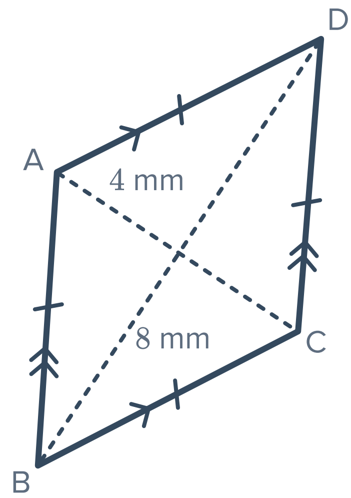
Areas of Kites and Rhombuses

20 Find the area of each rhombus below:

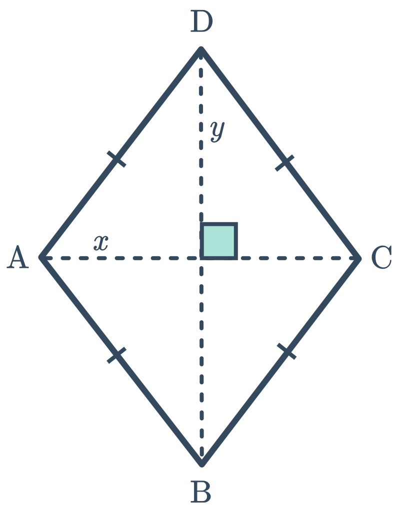
a



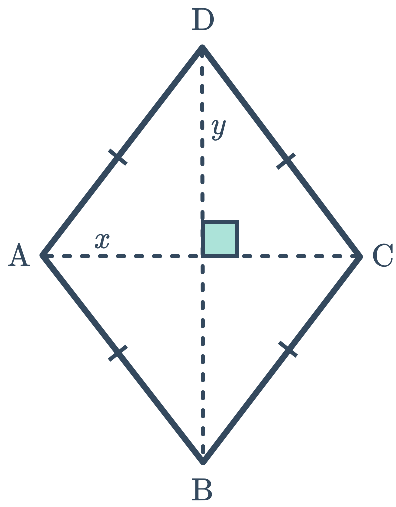
b



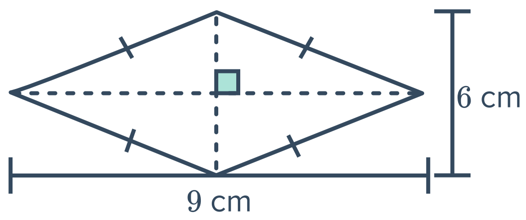
c $x = 10$ and $y = 12$



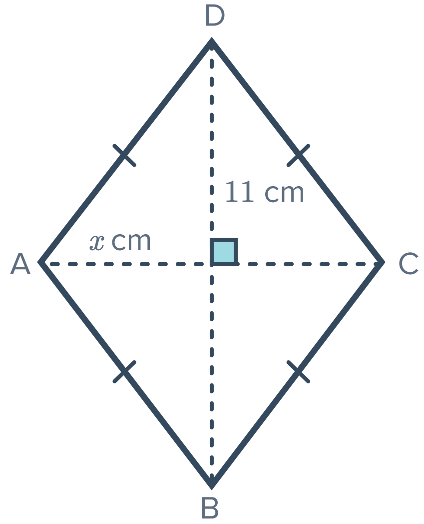
d $x = 4\,\text{cm}$ and $y = 8\,\text{cm}$



e

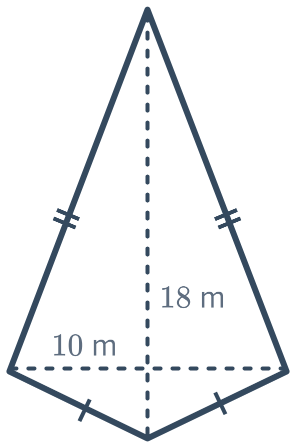


21 Rhombus $ABCD$ has an area of $55\,\text{cm}^2$. If diagonal $BD = 11\,\text{cm}$, and $AC = x\,\text{cm}$, find x.

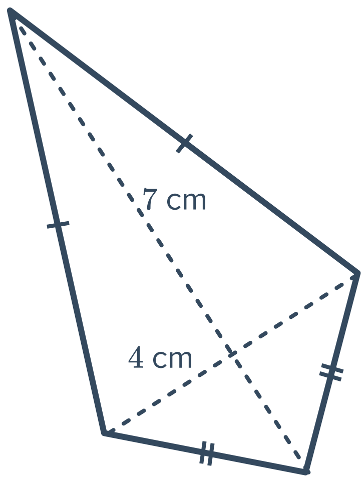


22 Find the area of the following kites:

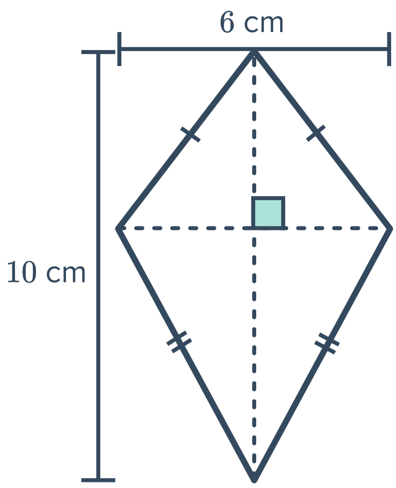
a



b

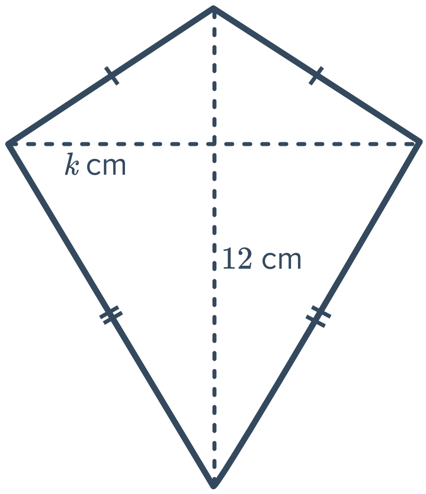


c



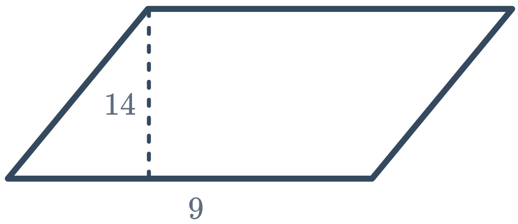
23 The area of a kite is $308\ \text{cm}^2$ and one of th[redacted]gonals is 47 cm. If the length of the other diagonal is y cm, what is the exact value of y?

24 The following kite has an area of $48\,\text{cm}^2$.
The length of one of its diagonals is $12\,\text{cm}$.
Find the length of the other diagonal, k.

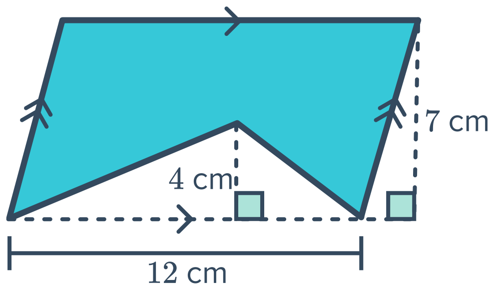


Applications

25 A rectangular sign casts a shadow on the ground as shown. What is the area of the shadow?

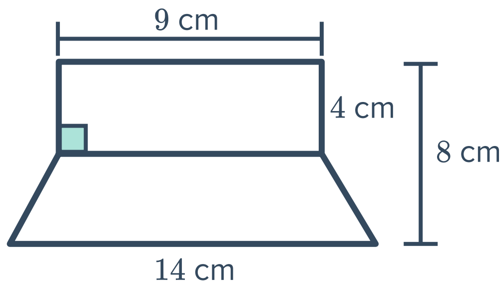


26 Find the shaded area shown in the figure:

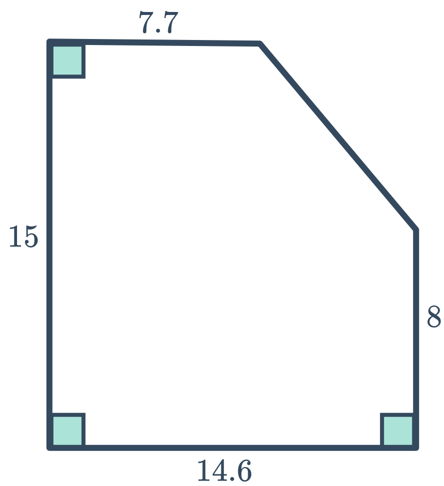


27 Sharon has purchased a rectangular piece of fabric measuring 12 m in length and 7 m in width. What is the area of the largest triangular piece she can cut out from it?

28 A rectangle is placed on top of a trapezium as shown. Find the area of the composite shape.



29 Find the area of the figure shown. All dimensions are in millimetres.



dimensions are in metres.